

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 35)

Suggested Solution

Paper 1

- | | | |
|-------|-------|-------|
| 1. C | 11. D | 21. D |
| 2. B | 12. C | 22. D |
| 3. C | 13. B | 23. C |
| 4. C | 14. D | 24. A |
| 5. D | 15. D | 25. A |
| 6. B | 16. A | 26. A |
| 7. D | 17. A | 27. B |
| 8. B | 18. C | 28. A |
| 9. B | 19. D | 29. C |
| 10. B | 20. A | 30. A |

Explanation

1. Answer: C

The time that Anson has already spent on studying economics is a cost that has been paid in the past and cannot be recovered, which is not the opportunity cost of the choice. It does not affect current decision making.

2. Answer: B

Mr. Chan's cost of continuing to live in the residential unit is the value of selling the residential unit and the monetary cost of continuing to live in the residential unit (i.e. the renovation cost and the rent paid to rent a residential unit during the renovation period).

Option B is correct. If the renovation work is delayed, Mr. Chan will have to rent another residential unit to live in for a longer period, which will increase his monetary cost of continuing to live in the residential unit.

Option A is incorrect. Mr. Chan has to give up the income that could have been obtained from selling the residential unit. If the market price of the residential unit decreases, his cost of continuing to live in the residential unit will decrease.

Option C is incorrect. If the market rent of other residential units decreases, the monetary cost of continuing to live in the residential unit will decrease.

Option D is incorrect. If the quality of the renovation work turns out to be better than expected, the value of continuing to live in the residential unit will increase, but the highest-valued option forgone will not be affected.

3. Answer: C

Option A is incorrect. Even if a good is scarce, if the market price is at equilibrium and the quantity demanded equals the quantity supplied, there will be no shortage of the good.

Option B is incorrect. If a good is scarce, there is a cost involved in obtaining the good. However, it can be free of charge.

Option D is incorrect. Scarcity is a relative concept. A good is scarce when its quantity available is not sufficient to satisfy all human wants, regardless of whether its quantity is fixed.

4. Answer: C

5. Answer: D

Option A is incorrect. The delivery services are producer goods, but the explanation in the option does not match the definition of producer goods.

Option B is incorrect. Goods that are free of charge are not necessarily free goods. Economic goods may also be provided free of charge to consumers.

Option C is incorrect. The delivery services are not public goods because they are excludable and rival in consumption.

6. Answer: B

Option D is incorrect. Public goods are goods which are both non-rival and non-excludable in consumption. There is no direct relationship between the type of economic system and the quantity of public goods. The quantity of public goods may not increase.

7. Answer: D

The decision regarding which learning software to install on the tablet computers is related to the kinds of resources used to provide educational services. It is related to the 'how to produce' question.

The arrangement in which all Form 1 students are automatically included in the scheme is about who can receive the service. It is related to the 'for whom to produce' question.

8. Answer: B

(1) is correct. A private limited company is a legal entity, which means that Jonathan can enjoy limited liability.

(3) is correct. A private limited company can issue shares and bonds to raise capital while a partnership cannot, so a private limited company has a wider source of capital.

(2) is incorrect. The ownership of both a private limited company and a partnership can only be transferred with the consent of other owners.

9. Answer: B

10. Answer: B

11. Answer: D

12. Answer: C

(3) is incorrect. Cashiers are regarded as labour of a supermarket.

13. Answer: B

	Firm A	Firm B
Labour supply	$10 \text{ hours} \times 40 = 400 \text{ man-hours}$	$7 \text{ hours} \times 50 = 350 \text{ man-hours}$
Average labour productivity	$4,000 \text{ units} / 400 \text{ man-hours}$ $= 10 \text{ units per man-hour}$	$4,200 \text{ units} / 350 \text{ man-hours}$ $= 12 \text{ units per man-hour}$

According to the above table, firm A has a larger labour supply than firm B, while firm B has a higher average labour productivity than firm A.

14. Answer: D

As firm A produces the same amount of output as firm B with more man-hours, firm A has a lower labour productivity than firm B (firm B has a higher labour productivity than firm A).

Option B is incorrect. If firm A adopts piece rate while firm B adopts time rate to pay their workers, as the incomes of firm A's workers directly depend on output, the labour productivity of firm A will likely be higher than that of firm B, which is inconsistent with the information given in the question.

15. Answer: D

Increasing the subsidy ceiling of the Continuing Education Fund can help workers acquire more skills needed for certain jobs, increasing their ability to change from one occupation to another. The occupational mobility of labour will thus increase.

Option A is incorrect. When more industries implement a licensing system, it will become more difficult for people to enter those industries, thus reducing the occupational mobility of labour.

Option B is incorrect. Adopting a higher degree of division of labour will lower the occupational mobility of labour as workers will lack skills required by other jobs.

Option C is incorrect. Increasing the monthly subsidy under the Public Transport Fare Subsidy Scheme will reduce transportation costs for workers. The geographical mobility of labour will thus increase.

16. Answer: A

Machinery (units)	Labour (units)	Total product of labour (units)	Average product of labour (units)	Marginal product of labour (units)
10	1	5	5	5
10	2	12	6	7
10	3	24	8	12
10	4	32	8	8
10	5	35	7	3

(1) is correct. When the quantity of labour increases from 3 units to 4 units, the marginal product of labour decreases from 12 units to 8 units.

(2) is correct. As there is a fixed factor (machinery), the information shows a short run situation.

17. Answer: A

Total output (units)	Total cost (\$)	Average cost (\$)
1,000	5,000	5
2,000	8,000	4

(1) is correct. Average cost decreases when the production scale increases, thus the firm enjoys internal economies of scale.

(2) is incorrect. The law of diminishing marginal returns only applies in the short run.

(4) is incorrect. The effect of increasing output on the firm's profit is uncertain. We do not know whether the firm will be able to sell all the additional output. Also, we do not know what the selling price will be when the firm expands its output.

18. Answer: C

When there are more taekwondo schools in the market, the market supply of taekwondo courses will increase and the market price will decrease. Marginal revenue will decrease and the profit-maximising output will decrease.

Option A is incorrect. When the monthly salary of taekwondo instructors decreases, the marginal cost of production will decrease. The profit-maximising output will increase.

Option B is incorrect. When an increasing number of primary schools place more importance on students' extracurricular performance in sports in admission decisions, demand for taekwondo courses will increase and the market price will increase. Marginal revenue will increase and the profit-maximising output will increase.

Option D is incorrect. When the rent for the premises of Mr. Wong's taekwondo school decreases, the fixed cost of production will decrease. The profit-maximising output will not be affected.

19. Answer: D

The manufacturer of dairy products expanded into a previous stage of production.

Option A is an example of vertical forward integration. The food wholesale company expanded into a latter stage of production.

Option B is an example of lateral integration. The hotel expanded into the production of related but not directly competing services.

Option C is an example of conglomerate integration. The bakery expanded into the production of totally different goods.

20. Answer: A

When deriving the market demand curve of a good, economists hold all the factors affecting a consumer's quantity demanded other than its price constant to study the effect of changes in price on the quantity demanded. Therefore, the price of the good and its quantity demanded are not assumed to be constant.

21. Answer: D

(1) is correct. Some people were willing to pay the price but still could not buy the packages, which means that at the prevailing price, the quantity demanded was greater than the quantity supplied. The quantity demanded is greater than the quantity supplied when the market price is lower than the equilibrium price.

(2) is correct. As the quantity demanded was greater than the quantity supplied at the prevailing price, there was a shortage.

(3) is correct. In addition to the price of the packages, people also needed to wait in line on the platform. The opportunity cost of obtaining a package included not only the package price, but also the time cost involved in buying the package.

22. Answer: D

The money price of high-quality furniture is usually higher than that of average-quality furniture. After the lump-sum delivery fee is raised, the relative price of high-quality furniture (in terms of average-quality furniture) will decrease. According to the law of demand, people will buy more high-quality furniture relative to average-quality furniture. The proportion of high-quality furniture sold in the market will increase.

Before the delivery fee is raised, the ratio of high-quality furniture sold to average-quality furniture sold is $12,000:8,000 = 1:0.67$.

(2) is correct. The ratio of high-quality furniture sold to average-quality furniture sold is $10,000:7,500 = 1:0.75$. The proportion of high-quality furniture sold in the market decreases, which is inconsistent with the law of demand.

(3) is correct. The ratio of high-quality furniture sold to average-quality furniture sold is $10,000:7,000 = 1:0.7$. The proportion of high-quality furniture sold in the market decreases, which is inconsistent with the law of demand.

(1) is incorrect. The ratio of high-quality furniture sold to average-quality furniture sold is $11,500:7,500 = 1:0.65$. The proportion of high-quality furniture sold in the market increases, which is consistent with the law of demand.

23. Answer: C

Since a fixed transportation cost is incurred in transporting fruits to a faraway market, the relative price of low-quality fruits will increase. According to the law of demand, people shopping in a faraway market will buy relatively fewer low-quality fruits than people shopping in a market near the farm.

Option A is incorrect. Charging a 10% service fee will not affect the relative prices of deluxe rooms and standard rooms. Therefore, the ratio of people choosing deluxe rooms to standard rooms will not change.

Option B is incorrect. When the fixed penalty for littering increases, the cost of littering will increase. According to the law of demand, the number of people littering will decrease.

Option D is incorrect. When a per-unit import tariff is imposed on red wine, the relative price of high-quality red wine will decrease. According to the law of demand, relatively more high-quality red wine will be imported. The average quality of imported red wine will improve.

24. Answer: A

Gasoline cars and electric cars are substitutes. When the price of gasoline cars increases, the demand for electric cars will increase.

Option B is incorrect. When the interest rate on car loans increases, some people may be unable or unwilling to borrow at a higher interest rate. The demand for electric cars will decrease.

Option C is incorrect. When people have safety concerns about electric cars, their demand for electric cars will decrease.

Option D is incorrect. When the price of new electric cars decreases, the quantity demanded, not demand, will increase.

25. Answer: A

When more people keep pets at home, the demand for pet care services will increase. The demand curve will shift to the right from D_0 to D_1 .

When more young people take animal care courses and join the pet care industry, the supply of pet care services will increase. The supply curve will shift to the right from S_0 to S_1 .

26. Answer: A

An increase in the demand for good X will lead to an increase in the price of good X.

When good X and good Y are in joint supply, an increase in the price of good X will lead to an increase in the supply of good Y. The price of good Y will decrease and the quantity demanded will increase.

Options C and D are incorrect. When good X and good Y are in competitive supply, an increase in the price of good X will lead to a decrease in the supply of good Y. The price of good Y will increase and the quantity demanded will decrease.

27. Answer: B

The completion of private residential properties will lead to an increase in the supply of private rental residential units. When more people hold a pessimistic view of future property prices, they will choose to rent a residential unit instead of buying one. Demand for the private rental residential units will increase.

28. Answer: A

When the price increases, Miss Chan's total expenditure increases while her quantity demanded decreases, indicating that the percentage increase in price is greater than the percentage decrease in quantity demanded. Her demand for fitness services is inelastic. The value of the price elasticity of demand coefficient is greater than 0 and smaller than 1.

When the price increases, Mr. Tong's quantity demanded remains unchanged (20 units), indicating that his demand for fitness services is perfectly inelastic. The value of the price elasticity of demand coefficient is 0.

29. Answer: C

An increase in the reproductive rate of cows will lead to an increase in the supply of local beef, causing the price to decrease and the quantity transacted to increase.

(2) is correct. Since the percentage decrease in the price of local beef (40%) is smaller than the percentage increase in the quantity transacted (55%), the total expenditure on local beef will increase.

(3) is correct. Local beef and imported beef are substitutes. A drop in the price of local beef will reduce the demand for imported beef. The quantity transacted of imported beef will decrease.

(1) is incorrect. An increase in the supply of local beef will cause the quantity demanded of local beef to increase, not the demand.

(4) is incorrect. A decrease in the demand for imported beef will cause the price and quantity transacted to decrease. The total expenditure on imported beef will decrease.

30. Answer: A

When people's income in general increases, demand for nutritional supplements will increase. The price and quantity transacted will increase. As the percentage change in price (15%) is smaller than the percentage change in quantity transacted (20%), the supply of nutritional supplements is elastic.

Option A is correct. If the nutritional supplement manufacturers have spare production capacity, the supply of nutritional supplements tends to be more elastic. When the price increases, they can increase output flexibly by making use of their spare production capacity.

Option C is incorrect. If there are strict licensing requirements for nutritional supplement manufacturers, new manufacturers cannot enter the industry easily. The supply tends to be less elastic.

Options B and D are incorrect. They are factors affecting price elasticity of demand.

Paper 2

Section A (30 marks)

1. (a) The opportunity cost will not change. (1 mark)

This is because this will only decrease the value of Karson's chosen option without affecting his highest-valued option forgone. (1 mark)

- (b) The change in the opportunity cost will be uncertain. (1 mark)

The tutoring centre raising the wage for part-time tutors will increase the value of the third option. If the value of the third option does not exceed the value of the second option, the cost of choosing to travel around the United Kingdom will not change. If the value of the third option increases to a level higher than that of the second option, working as a part-time tutor at a tutoring centre will become the highest-valued option forgone. The cost of choosing to travel around the United Kingdom will change from working as a summer volunteer at a community centre to working as a part-time tutor in a tutoring centre. (2 marks)

2. (a) No. (1 mark)

The broadcasts are produced from scarce resources which have alternative uses. (1 mark)

- (b) Yes. (1 mark)

The broadcasts are non-rival in consumption. They can be consumed by many individuals concurrently without reducing the amount available for others. (2 marks)

3. (a) The quantity of machinery does not change with output. (1 mark)

(b)

Labour (units)	Machinery (units)	Average product of labour (units)	Total product (units)	Marginal product (units)
10	5	5	50	-
11	5	7	77	27
12	5	10	120	43
13	5	12	156	36
14	5	12	168	12

Yes. Given the fixed quantity of machinery (5 units), when labour increases from 12 to 13 units, the marginal product of labour drops from 43 to 36 units, so the data illustrate the law of diminishing marginal returns. (3 marks)

4. (a) An example of a fixed cost:

- rental of the shop
- business registration fee
- any reasonable answer

(Mark the **FIRST** point only)

1 @, max: 1 mark

- (b) An advantage:

- The procedure of setting up a sole proprietorship is simple.
- A sole proprietorship is subject to a lower profits tax rate.
- Any reasonable answer

(Mark the **FIRST** point only)

1 @, max: 1 mark

A disadvantage:

- A sole proprietor bears unlimited liability.
- A sole proprietorship has limited continuity.
- Any reasonable answer

} 1 @, max: 1 mark

(Mark the **FIRST** point only)

(c) Verbal elaboration:

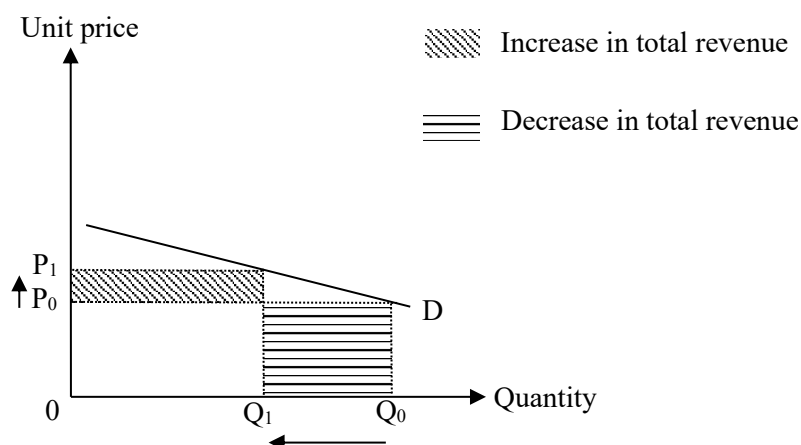
If the demand for the haircut services is elastic, the percentage increase in price will be smaller than the percentage decrease in quantity demanded, and total revenue will therefore decrease.

(3 marks)

Illustrate in the diagram:

- correct positions of the increase in total revenue and the decrease in total revenue (1 mark)

- decrease in total revenue > increase in total revenue (1 mark)



5. (a) Tertiary production. (1 mark)

This is because hotels provide accommodation services. (1 mark)

(b) When a fixed administrative fee is charged, the relative price of longer stays (in terms of shorter stays) will decrease. According to the law of demand, a higher proportion of guests will choose to stay longer. As a result, guests' average length of stay will increase. (4 marks)

(c) The guest does not have the right to transfer the rented room, as he cannot sell the room. (2 marks)
OR

The guest does not have the exclusive right to receive income from the room as he cannot rent the room out to generate income. (2 marks)

OR

Any reasonable answer (2 marks)

Section B (40 marks)

6. (a) (i) Complex division of labour. (1 mark)
Workers specialise in a particular stage of production in the restaurant. (1 mark)

(ii) Advantages:

- increased (average) labour productivity
- economy in the use of capital
- any reasonable answers

1@, max: 2 marks

(Mark the **FIRST TWO** points only)

(b) An example of a variable factor:

- part-time workers
- ingredients
- any reasonable answer

1@, max: 1 mark

(Mark the **FIRST** point only)

(c) An advantage:

- The cost of measuring workers' output is lower because wages are paid on a time basis.
- Since workers' incomes are more stable, it is easier to maintain a static team.
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

A disadvantage:

- Since workers' incomes do not directly depend on output, they tend to be less hard-working and have weaker incentive to work hard.
- The cost of supervising workers' performance is higher because of the problem of shirking.
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

7. (a)

Output (units)	1	2	3	4	5
Average cost (\$)	10	12	14	16	18
Total cost (\$)	10	24	42	64	90
Marginal cost (\$)	-	14	18	22	26
Marginal revenue (\$)	18	18	18	18	18

The profit-maximising output level is 3 units. At this level, marginal revenue = marginal cost = \$18.

(1 mark)

The corresponding profit is \$12 (= \$18 × 3 – \$42).

(1 mark)

(b)

Output (units)	1	2	3	4	5
Total cost (\$)	6	16	30	48	70
Marginal cost (\$)	-	10	14	18	22
Marginal revenue (\$)	18	18	18	18	18

The profit-maximising output level increases from 3 units to 4 units. At this level, marginal revenue = marginal cost = \$18. (2 marks)

The corresponding profit is \$24 ($= \$18 \times 4 - \48), which is an increase of \$12 ($= \$24 - \12).

(1 mark)

(c) (i) Vertical forward integration.

(1 mark)

This is because SWIM Well has expanded its operations vertically forward to a latter stage of production.

(1 mark)

(ii) Advantages:

- to ensure a market outlet for its products
- to enjoy economies of scale
- to use resources more efficiently
- to take advantage of brand names
- any reasonable answers

1@, max: 2 marks

(Mark the **FIRST TWO** points only)

8. (a) No.

Full cost = Monetary cost + Time cost

Even if the emergency dental treatment services are provided free of charge, the time cost of queuing for the services may not be the same for everyone, as they may have different alternative uses for their time. (3 marks)

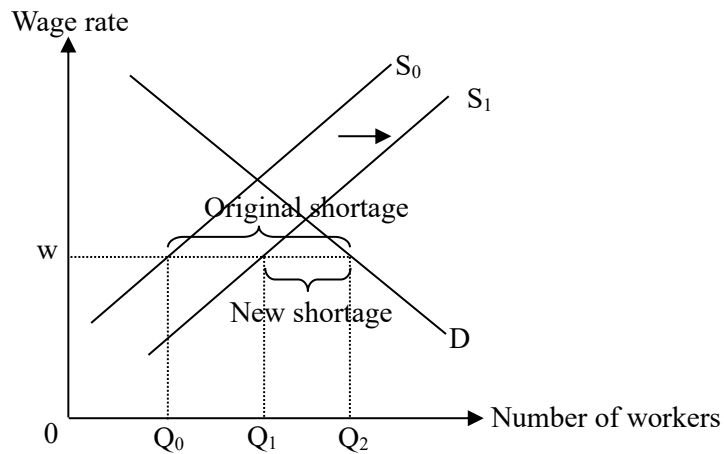
(b) Verbal elaboration:

Initially the wage rate is below the equilibrium level, so the quantity demanded of dentists is greater than the quantity supplied. There is excess demand (a shortage) in the market.

When a new dental school is established, the supply of dentists will increase. Given the wage rate remains unchanged, an increase in the supply will lead to a decrease in the excess demand for dentists. (3 marks)

Illustrate in the diagram:

- the prevailing wage is below the equilibrium wage rate (1 mark)
- correct position of the original shortage (1 mark)
- rightward shift of supply curve (1 mark)
- correct position of the new shortage (a decrease in excess demand) (1 mark)



- (c) Positive statement. It is refutable by facts, i.e. people can use the data on the public dental market from before and after the adjustment of the remuneration of public dentists to verify whether the statement is correct. (2 marks)
- (d) The supply of dentists may be inelastic because they require special training and qualification. Even if the government raises the wage rate of public dentists, the percentage increase in the quantity supplied of dentists would still be smaller than the percentage increase in the wage rate. (3 marks)
- (e) Marks awarded for effective communication (EC: max 1 mark)

Marks	Performance
1	- Arguments supported with the source/data and appropriate economic theories - Relevant materials presented - Well-organised and coherent answers presented without repetition of ideas - Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
0	- Arguments presented with no support of the source/data and economic theories - Materials unrelated to the gist of the question presented - Inconsistent arguments presented - Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 6 marks. Answers may include the following:

Politicians who are concerned about whether low-income individuals have timely access to dental services

- Politicians who are concerned about whether low-income individuals have timely access to dental services would prefer Proposal I. Public dental services and private dental services are substitutes. Subsidising the use of private dental services will reduce the fees patients pay to use private dental services. Some patients with urgent medical needs may switch to private dental services as they can receive treatment at a lower price. As some patients switch to private dental services, the demand for public dental services will decrease. The shortage of public dental services will decrease, thus shortening the queuing time for patients who rely on public dental services, and they can receive treatment faster.
- Politicians who are concerned about whether low-income individuals have timely access to dental services would prefer Proposal II. Even if the government subsidises the use of private dental services, some low-income patients may still not be able to afford the medical expenses of private dental clinics and will not be able to use private dental services due to budget constraints. Hiring more non-locally trained dentists to provide public dental services can increase the supply of public dental services. The shortage of public dental services will decrease, thus shortening the queuing time for patients, and they can receive treatment faster.

Owners of private dental clinics

- Owners of private dental clinics would prefer Proposal I. Subsidising the use of private dental services will increase the demand for private dental services. Total revenue including the subsidy for private dental clinics will increase.
- Owners of private dental clinics would not prefer Proposal II. Public dental services and private dental services are substitutes. Hiring more non-locally trained dentists to provide public dental services will increase the supply of public dental services, thus reducing the queuing time for public dental services. Citizens' full cost of using public dental services will decrease, causing the demand for private dental services to decrease. The total revenue of private dental clinics will decrease.